Day 093 | Unit 07 | Teacher plan

### Understand and Use Percent

# Represent Percents Above 100

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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## 01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

## Worked example 1

Write 170% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 170% = 170/100 = 17/10.

4. Dividing by 100 gives 1.7, which exceeds 1 because 170 exceeds 100.

5. State the result in the form requested: 17/10; 1.7.

Answer: 17/10; 1.7

## Worked example 2

Select all representations of 375%.
A. 375
B. 375/100
C. 375/10
D. 3.75

1. Percent means per 100.

2. Percent means per 100.

3. Divide 375 by 100 to get 3.75

4. simplify 375/100 to 15/4.

Answer: B, D - 375/100; 3.75

# Day 093 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

Write 150% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

### 02 / Constructed response

Write 210% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

### 03 / Constructed response

Write 155% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

### 04 / Constructed response

Write 110% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

### 05 / Constructed response

Write 290% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

### 06 / Constructed response

Write 235% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

## 07 / Constructed response

Write 160% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

## 08 / Multiple select

Select all representations of 275%.
A. 275
B. 275/100
C. 2.75
D. 275/10

## 09 / Multiple select

Select all representations of 325%.
A. 325/100
B. 325/10
C. 325
D. 3.25

## 10 / Constructed response

Write 115% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

Day 093 | Unit 07 | Answer key

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### 1. 3/2; 1.5

150% = 150/100 = 3/2. Dividing by 100 gives 1.5, which exceeds 1 because 150 exceeds 100.

6.RP.A.3c | Constructed response

### 2. 21/10; 2.1

210% = 210/100 = 21/10. Dividing by 100 gives 2.1, which exceeds 1 because 210 exceeds 100.

6.RP.A.3c | Constructed response

### 3. 31/20; 1.55

155% = 155/100 = 31/20. Dividing by 100 gives 1.55, which exceeds 1 because 155 exceeds 100.

6.RP.A.3c | Constructed response

### 4. 11/10; 1.1

110% = 110/100 = 11/10. Dividing by 100 gives 1.1, which exceeds 1 because 110 exceeds 100.

6.RP.A.3c | Constructed response

### 5. 29/10; 2.9

290% = 290/100 = 29/10. Dividing by 100 gives 2.9, which exceeds 1 because 290 exceeds 100.

6.RP.A.3c | Constructed response

### 6. 47/20; 2.35

235% = 235/100 = 47/20. Dividing by 100 gives 2.35, which exceeds 1 because 235 exceeds 100.

6.RP.A.3c | Constructed response

### 7. 8/5; 1.6

160% = 160/100 = 8/5. Dividing by 100 gives 1.6, which exceeds 1 because 160 exceeds 100.

6.RP.A.3c | Constructed response

## 8. B, C - 275/100; 2.75

Percent means per 100. Divide 275 by 100 to get 2.75; simplify 275/100 to 11/4.

6.RP.A.3c | Multiple select

## 9. A, D - 325/100; 3.25

Percent means per 100. Divide 325 by 100 to get 3.25; simplify 325/100 to 13/4.

6.RP.A.3c | Multiple select

## 10. 23/20; 1.15

115% = 115/100 = 23/20. Dividing by 100 gives 1.15, which exceeds 1 because 115 exceeds 100.

6.RP.A.3c | Constructed response